

Details	https://m.facebook.com/story.php?story_fbid=3078489702425349&id=100007930489461				
Join this discord server to enroll	https://discord.gg/z56rQ4xaqh				
Title	Phase 1 (8 sessions, 4 weeks)	Phase 2 (8 sessions, 4 weeks)	Phase 3 (8 sessions, 4 weeks)	Phase 4 (8 sessions, 4 weeks)	
Prerequisites	- You know basic C/C++ including variables, operators, conditions, functions and loops. If you don't know then learn it from Google. For example from https://www.javatpoint.com/cpp-tutorial or https://www.w3schools.com/CPP/default.asp or from any of your favorite youtube channels. - You are willing to spend at least 20 hours per week in CP and to participate in all major online contests.	- You know everything from Phase 1 - You are willing to spend at least 20 hours per week in CP and to participate in all major online contests.	- You know everything from Phase 1 and 2 - You are willing to spend at least 20 hours per week in CP and to participate in all major online contests.	- You know everything from Phase 1 and 2 - You are willing to spend at least 20 hours per week in CP and to participate in all major online contests.	
Introduction to Competitive Programming(CP)	- What the heck is CP? - Why bother to learn it when I can play a new game with that time? - What is a problem? - What is meant by AC, WA, TLE... etc? - Where to write codes? What are the best IDEs? - How do contests work? - What are some famous online judges? - How do Codeforces' contests work? - Can you show me a live demo by solving a problem because to me it seems like a rocket science? - What is ICPC? - What if I can't be very good at CP? - QnA				
Warm-up	- The OP header - Complexity Analysis(Time and Memory). - Int Overflows - Intro to arrays, subarrays and subsequences. - Intro to strings, substrings and subsequences. - Binary Number/Bitwise Operations - Huguieee input, what can be done? Intro to Fast I/O - Builtin Functions(reverse, swap, gcd, next_permutation, max , min, pow , powl, memset)				
Basic Problems	- Reverse of an array/a string - Sum of an array - Sum of digits of a number - Check if a string is palindrome or not - Check if two strings are anagram or not - Maximum and minimum of an array - Most frequent element of an array - Swap two variables without using a third variable - Prefix Sum - More easy practice problems				
Saviors	- Vector - List - Stack - Queue - Deque - Priority Queue - The Art of Debugging				
More Saviors	- Set, Unordered Set and Multiset - Map, Unordered Map and Gp Hash Table - Bitset - Multidimensional arrays - Order statistics tree and the number of inversions.				
Searching and Sorting	- Linear Search, Binary Search, Lower Bound, Upper Bound - Bubble Sort - Insertion Sort - Counting Sort - Merge Sort - C++ sort and how C++ implemented it				
Greedy	- Activity Selection Problem - Discussion of various greedy problems.				
Modular Arithmetic	- Intro - Bigmod - Mulmod - Modular inverse - Practice Problems				
Graph Theory		- Undirected/Directed Graphs, Trees, Types of graphs etc definitions - Adjacency Matrix and Lists - DFS, BFS - DFS Tree - Graph Coloring	- DSU - Dijkstra	- Floyd Warshall - MST - SCC, Articulation Points and Bridges	
Number Theory		- GCD and LCM - Sieve and factorization - Number of divisors - Euler Totient Funtion	- Segmented Sieve - Extended Euclid	- Inclusion Exclusion	
Math		- Basic math - Matrix Exponentiation	- Probabilities - Expected Value	(not math this time) - Bitmask DP - Digit DP	
DP		- Recursions - Intro to DP - 0/1 Knapsack - Longest Common Subsequence - DP with output tracking	- Kadane's Algorithm - LIS in $O(n^2)$ and $O(n \log n)$	- Tree DP	
Data Structures		- Segment Tree - Segment Tree Lazy and intro to BIT	- Sparse Table - LCA	- Sqrt Decomposition - MO's	
Strings		- KMP - Prefix Automaton - String Hashing	- Trie - Aho corasick	- Suffix Array	
Miscellaneous		- Ternary Search (on doubles and integers) - Coordinate Compression - Pigeonhole Principle	- Two Pointers - Sliding Window - Monotonous queue	- Game Theory - Grundy Numbers	
Soft Touch		- Combinatorics Basics	- Geometry Basics	- Difference array - Meet in the middle	
Additional	- Practice contests with curated problems. - Curated resources for each topic. - QnA or more like Adda after every session. - My phone number. - You can message me anytime with your problems and I will almost always reply instantly! - I will always be at your service to make you a better version of yourself.				